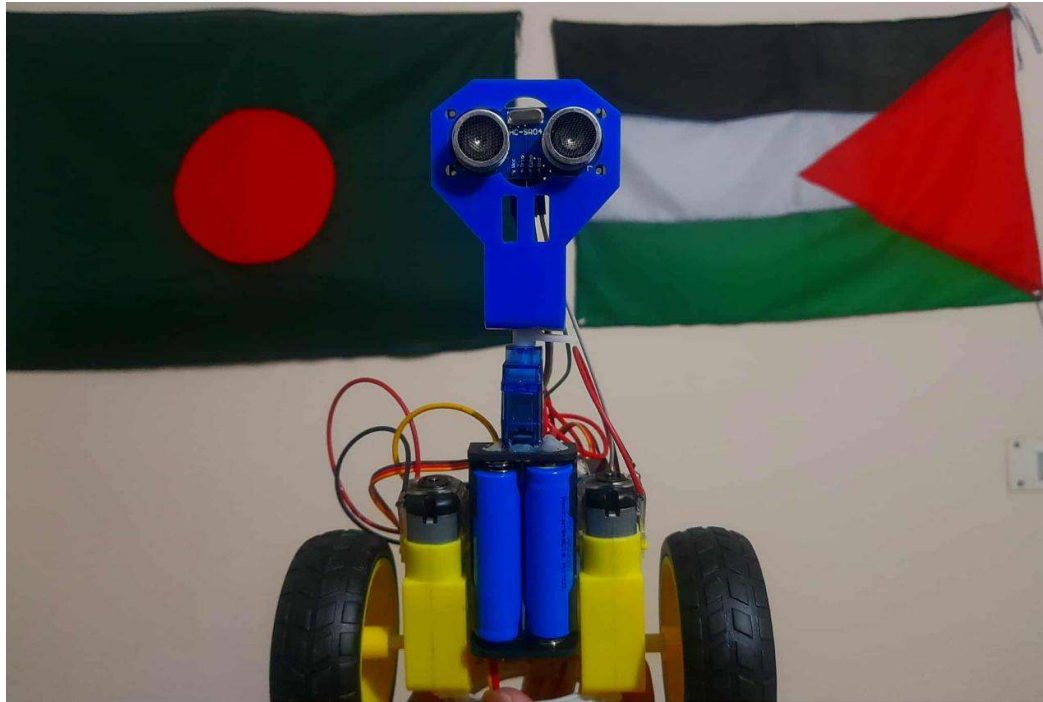


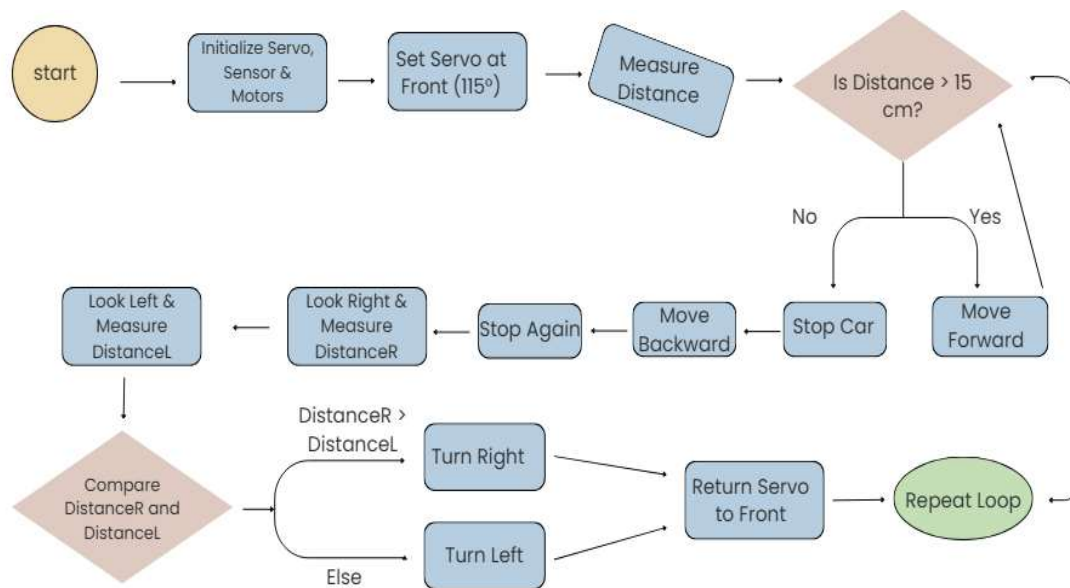


Obstacle-Avoiding Autonomous Car Robot Using Arduino

In this project, we will be introduced to **Arduino, Motor Shield, Servo Motor, and Ultrasonic Sensor**. Using these components, we will build an **Obstacle Avoiding Car** that can move autonomously, detect obstacles, and decide the safest path to continue. The project provides hands-on experience in **robotics, sensor integration, and motor control**, helping us understand the basics of autonomous navigation systems.

System Overview

This project demonstrates an **Arduino-based Obstacle Avoiding Car** that can detect obstacles in its path and automatically change direction to avoid collisions. The system uses an **ultrasonic sensor** mounted on a servo motor to scan the surroundings and a **motor driver shield** to control two DC motors.

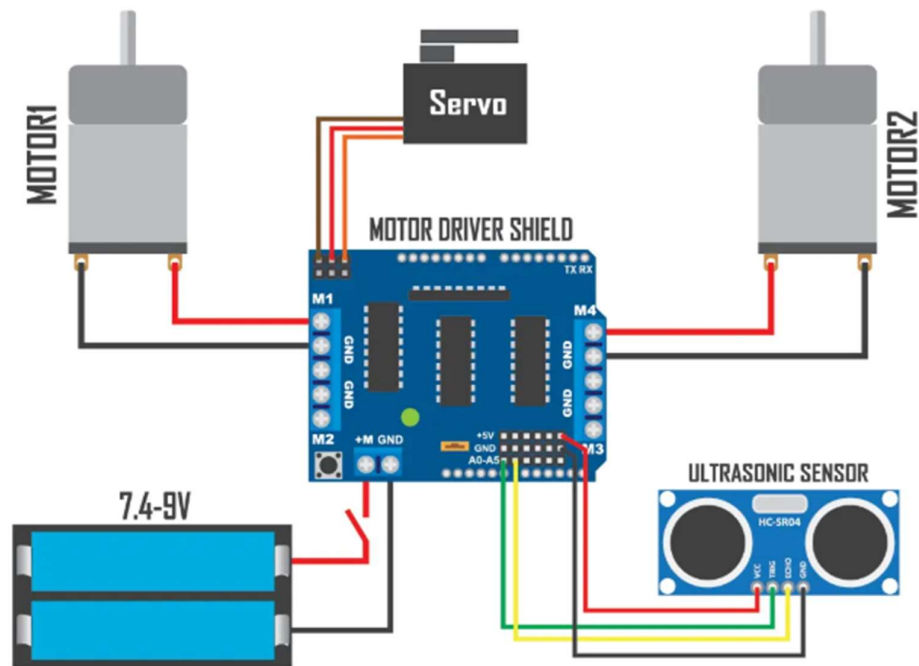


Flow Chart of The Project

Setup and Operational Guide

- 1) Arduino Uno
- 2) Arduino Motor Driver Shield
- 3) TT Gear Motor
- 4) Rubber Wheels
- 5) Servo Motor
- 6) Ultrasonic Sensor
- 7) 18650 Li-ion Battery
- 8) Battery Holder
- 9) Male and Female Jumper
- 10) DC Power Switch
- 11) Acrylic Sheet

Circuit Setup



How Does the System Work?

- The **ultrasonic sensor** continuously measures distance from obstacles.
- If an object is detected within **15 cm**, the car will stop, move backward slightly, and then scan left and right.
- The **servo motor** rotates the sensor to check distances on both sides.
- The car turns towards the side with more free space.
- If no obstacle is detected, the car keeps moving forward.

Useful Tools

- Arduino IDE (for programming)
- Jumper wires and breadboard (if prototyping)

- Multimeter (for debugging connections)

Summary

This project demonstrates a simple yet effective **Autonomous Obstacle Avoiding Car** using Arduino. The ultrasonic sensor and servo motor help in scanning the surroundings, while the motor driver shield controls the wheels to move forward, backward, or turn left/right. It is an excellent beginner-friendly project for learning about **robotics, automation, and sensor integration**.

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